SICSR SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)
(Established under Section 3 of the UGC Act, 1956) | Re-accredited by NAAC with 'A++' grade | Awarded Category – I by UGC

INDIAN RAIL PERFORMANCE: A 2025 DELAY SEVERITY STUDY

By

KARMANYA ANAND 25030140120

VARSHIL PATEL 25030140131

KOTTEDA GAURAV 25030140124

SHIVANG SRIVASTAVA 25030140147

Master of Business Administration In Information Technology

Symbiosis Institute of Computer Studies & Research

Symbiosis International (Deemed) University

PUNE – 411016

INTRODUCTION:

- The objective of this report is to analyze the Indian Railway Delay Dataset dataset using Power BI to derive meaningful insights for decision-making.
- The dataset underwent a thorough cleaning and transformation process to ensure accuracy and reliability in visualization and reporting.
- The key steps in data preparation included handling missing values, standardizing formats, and creating calculated fields for better analysis.
- This report presents the methodology, business relevance, schema design, dashboard visualizations, and insights that support strategic decision-making for Indian Railways.

BACKGROUND:

Indian Railways is one of the world's largest rail networks, serving as the backbone of India's transport infrastructure and a vital driver of the national economy. As a statutory body under the ownership of the Ministry of Railways, it operates thousands of passenger and freight trains daily, leveraging an extensive network of stations and divisions to facilitate the movement of millions of passengers and tons of cargo. With a massive operational footprint, Indian Railways relies increasingly on data analytics and digital transformation to manage complex scheduling, improve safety, and enhance the overall passenger experience.

Data analytics is essential for monitoring operational efficiency, tracking real-time train performance, and identifying systemic bottlenecks like frequent delays. By implementing a sophisticated data visualization and reporting solution in Power BI utilizing tables for train details, station information, and temporal calendars Indian Railways can gain actionable insights into punctuality trends and regional performance. This data-driven approach allows for optimized resource allocation and infrastructure planning, ultimately improving service reliability and driving modernization across the network.

DATA SOURCES:

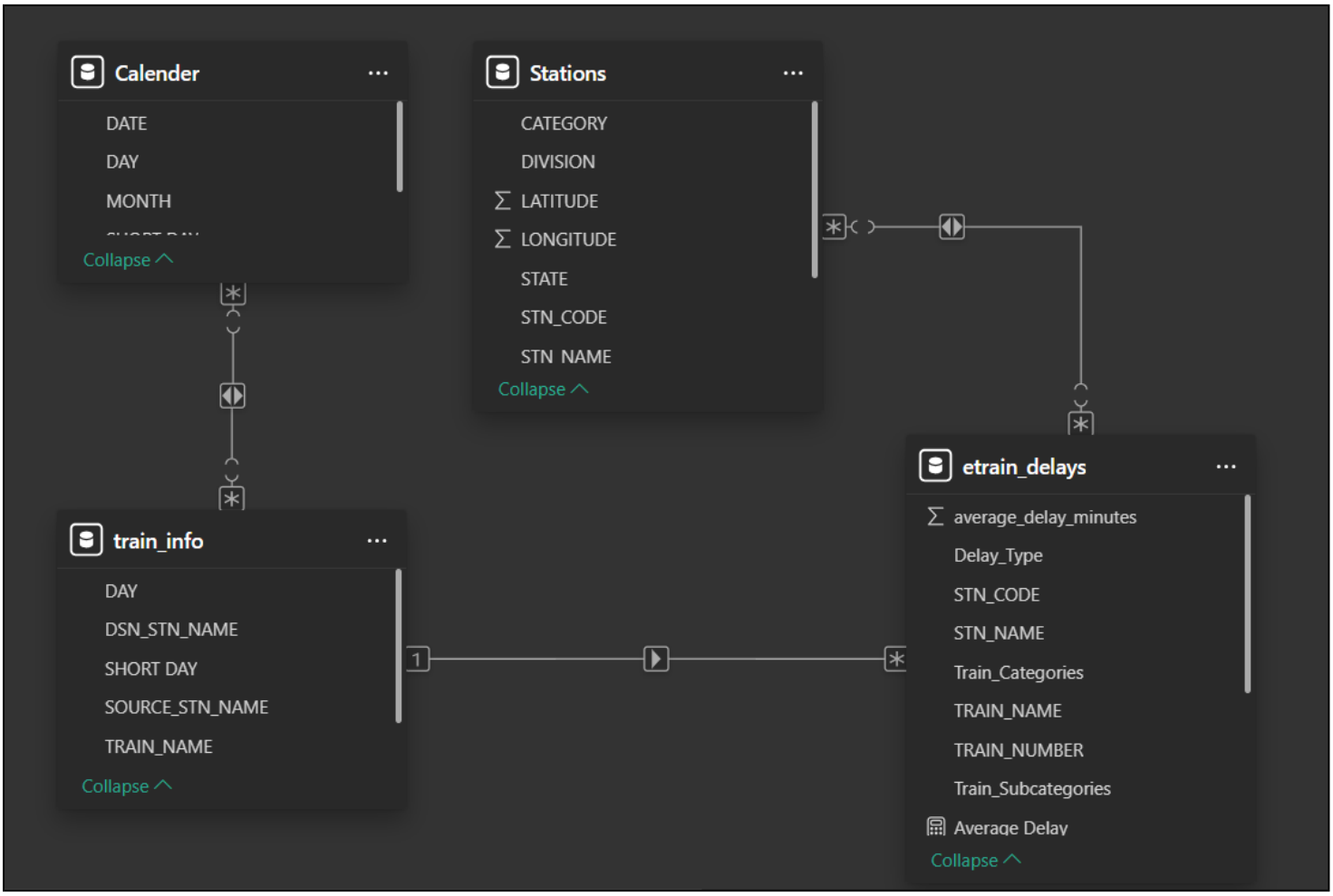
Dataset is collected from two different Kaggle datasets, one has information about all the trains and stations in India, and another has train delay dataset. All this data is cleaned and used for developing a dashboard.

Different measures and hierarchy’s have been created so that visualization can be shown in a more complex level.

A hierarchy between State and Division is created and a hierarchy between train categories and train subcategories has been created.

Average delay and on time train are two measures which had been created. Multiple new columns are created using conditional formatting for more forms of representation.

SCHEMA DIAGRAM:



A snowflake schema is applied in this model where `etrain_delays` is fact table and `train_info`, `Stations` are its dimensions.
`Calender` is a dimension branching out from `train_info`.

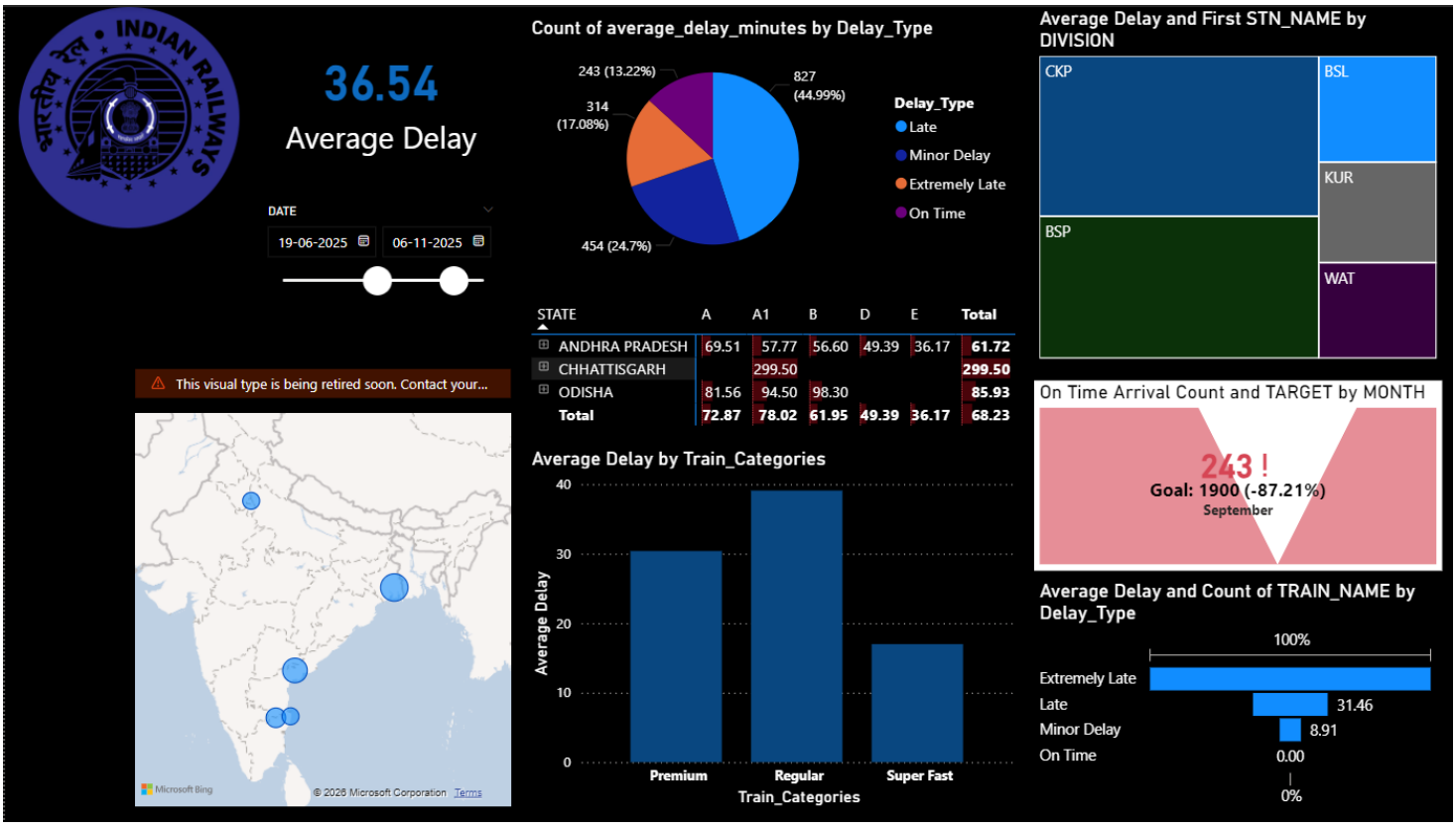
Normalized Dimensions: In a pure Star Schema, all descriptive data (like Month and Day) would be in a single `Calendar` table. In your model, you have a **chain**: `Calender -> train_info -> etrain_delays`.

Multiple Hops: To get from the `DATE` or `MONTH` in your Calendar to the actual delay data in `etrain_delays`, the relationship has to "pass through" the `train_info` table. This "branching out" of dimensions is the defining characteristic of a Snowflake.

Central Fact Table: Your `etrain_delays` table acts as the central **Fact Table**, containing the quantitative measures like `average_delay_minutes`.

DASHBOARD:

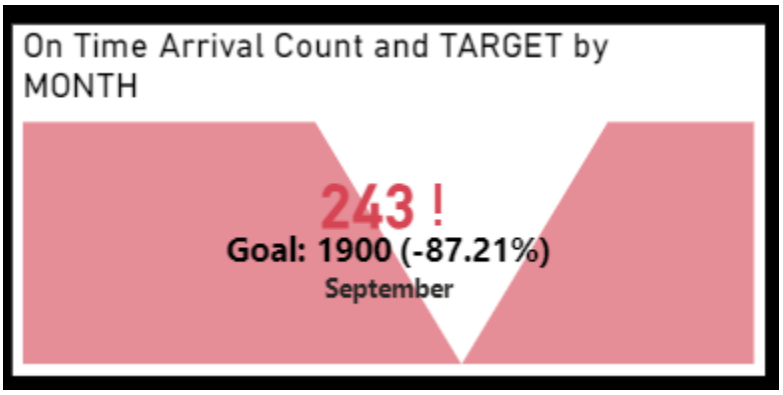
The Power BI dashboard presents a series of visualizations that facilitate data-driven decision-making. Key components include:



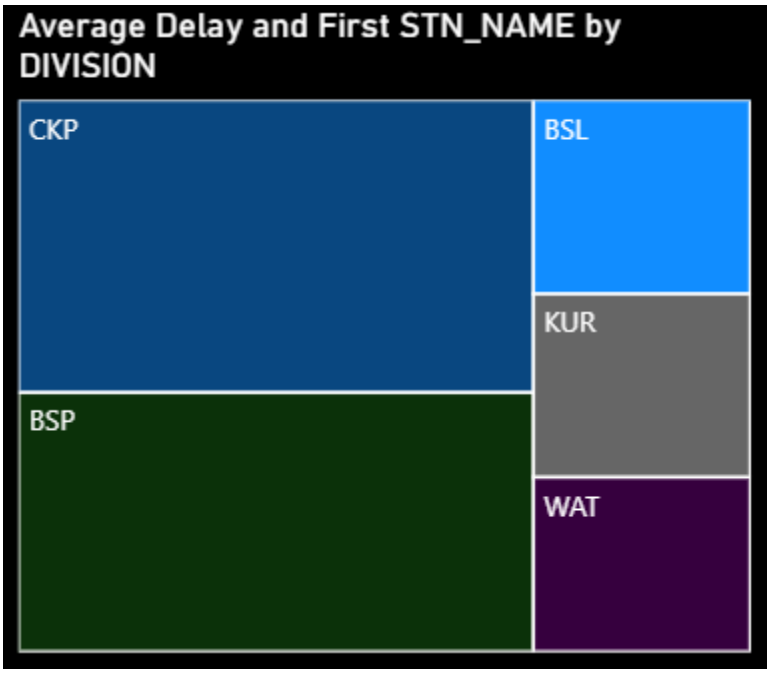
KPIs (key performance indicators)
This section provides a summary of essential business metrics:

Average Delay: 36.54 min, indicates the average delay of each train.

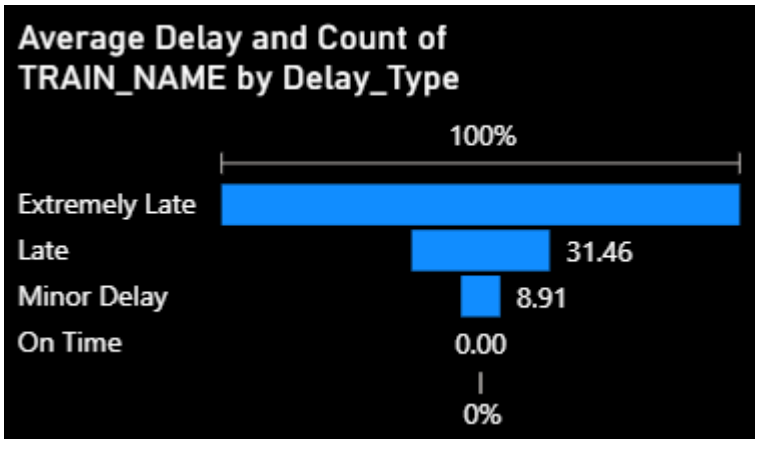
On time Arrival Count per Month: 243, represents the amount of trains arriving on time for the month September.
Goal represents total amount of trains.
-87.21 %, represents the amount of trains not arriving on time.
The red background represents the dip in number compared to previous month.
If the amount of on time arrival increases the background changes to green.



Average Delay by Division:
The treemap represents Stations with highest Average Delay for each Division
Due to hierarchy between State and Division, this visual can undergo drill up, drill down for state wise visual.

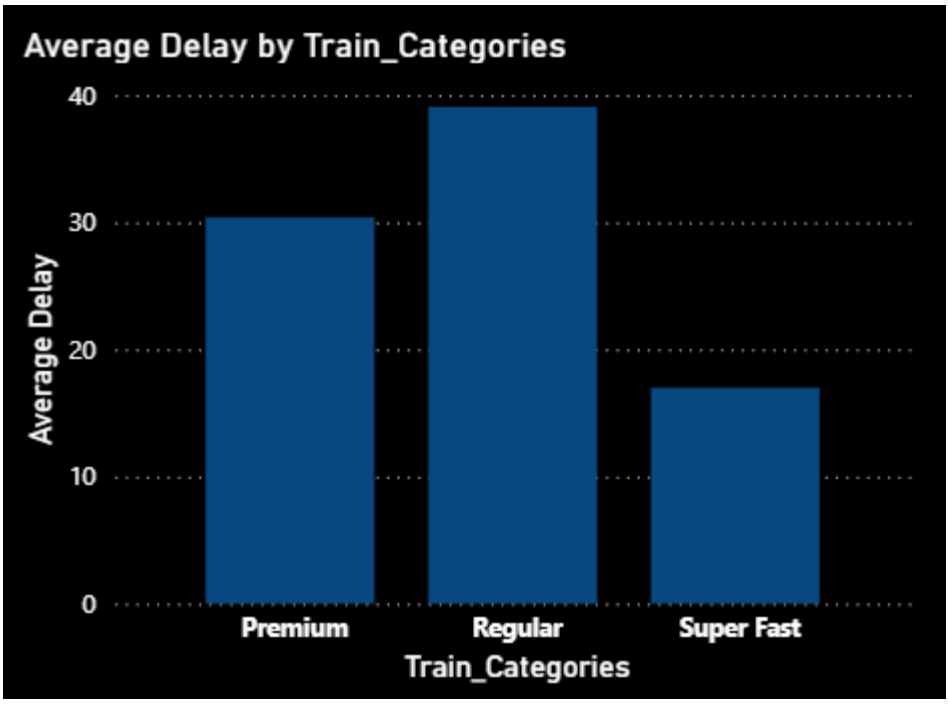


Number of Trains per Delay type:
This Funnel chart represents distribution of trains based on Delay type.



Average Delay by Train Categories:

Bar graph represents average delay by train categories. Drill/Drill down operation can be used to drill down train categories into train sub categories.



Total Delay By State:

This table shows the total delay for each state in different categorical stations.

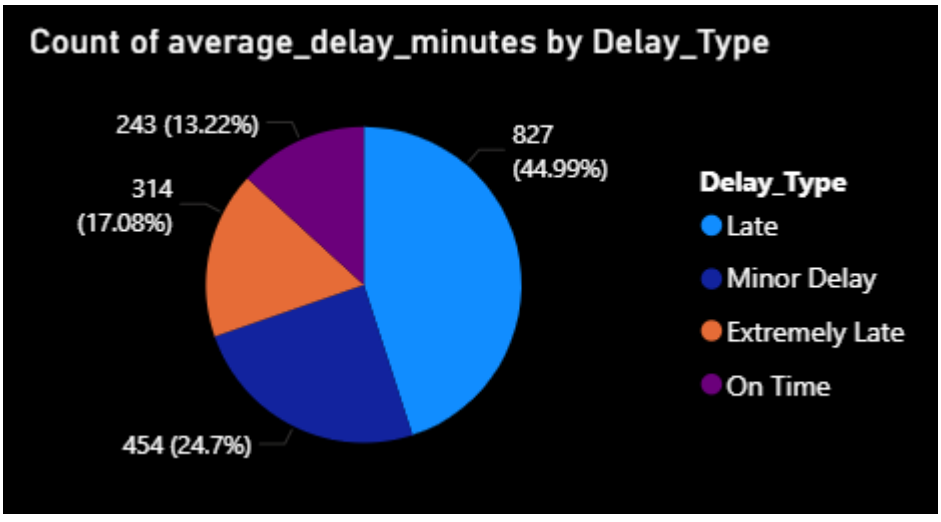
The red graph in each cell shows the amount of delay in graphical representation.

Drill down /Drill up operations can be performed on this table and divison wise total delay can be observed.

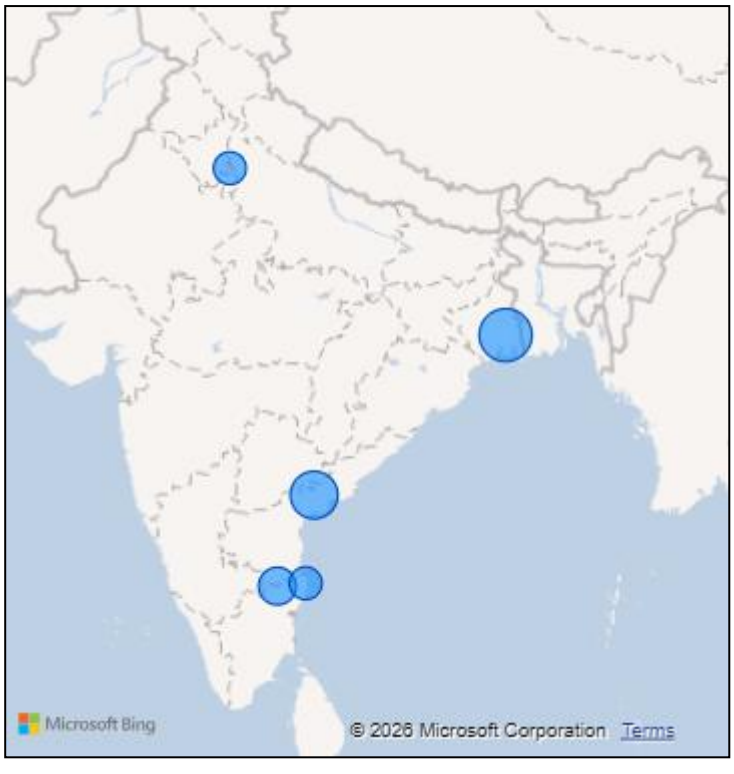
Top-3 filter is applied on this table to show the top 3 states with most delay.

STATE	A	A1	B	D	E	Total
CHHATTISGARH		299.50				299.50
ODISHA	81.56	94.50	98.30			85.93
ANDHRA PRADESH	69.51	57.77	56.60	49.39	36.17	61.72
Total	**72.87**	**78.02**	**61.95**	**49.39**	**36.17**	**68.23**

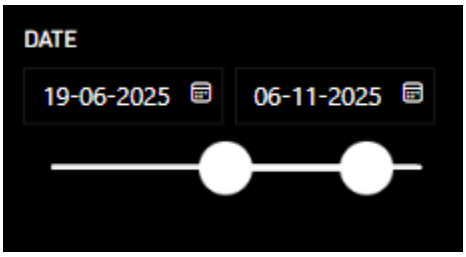
Number of Trains per Delay Type:
This pie chart categorizes the delay type into four different types and number of trains being categorized into them.



Map:
This visual represents the stations with most amount of delay.
The size of bubble is represented using delay amount. More the delay amount bigger the size of bubble.
A top-5 filter is applied on this visual based on average delay.



Slicer:
A slicer based on date is shown. Change in this visual is affected on every visual present in the dashboard.



CONCLUSION:

This Power BI project provides critical insights into the operational performance, punctuality trends, and regional bottlenecks of the Indian Railways network. The data architecture, which integrates station-level geographic data with detailed train delay logs, enables efficient reporting and supports strategic decision-making in areas such as resource allocation, infrastructure planning, and service modernization. By leveraging these insights to categorize delay severity and identify high-frequency failure points, Indian Railways can improve system-wide efficiency, reduce total minutes lost, and significantly enhance the passenger travel experience.